# Machine Learning - Q&A; Document

**Q:** What is machine learning?

**A:** Machine learning is a branch of artificial intelligence that enables systems to learn patterns from data and make predictions without being explicitly programmed.

**Q:** What are the types of machine learning?

**A:** The main types are supervised learning, unsupervised learning, and reinforcement learning.

**Q:** Define supervised learning.

**A:** Supervised learning is a method where the model is trained using labeled data, meaning input-output pairs are provided.

**Q:** What is unsupervised learning?

**A:** Unsupervised learning uses unlabeled data and finds hidden patterns or structures within it.

**Q:** Explain reinforcement learning.

**A:** Reinforcement learning is a type of learning where an agent learns by interacting with an environment and receiving rewards or penalties.

**Q:** What is overfitting?

**A:** Overfitting occurs when a model learns training data too well, including noise, and performs poorly on new data.

**Q:** What is underfitting?

**A:** Underfitting happens when a model is too simple to capture underlying patterns in the data.

**Q:** What are neural networks?

**A:** Neural networks are computing systems inspired by the human brain, consisting of layers of interconnected nodes.

**Q:** What is deep learning?

**A:** Deep learning is a subset of machine learning that uses multi-layered neural networks to learn complex patterns.

**Q:** What is the purpose of training data?

**A:** Training data is used to teach machine learning models by providing examples from which they learn patterns.